Industrial Equipment Anomaly & Incident Response Agent

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Contents

Overview **4**

Monitored Telemetry Parameters 4

System Architecture and Design **5**

Workflow Graph 5

Data Storage and Persistence Architecture 6

Simplified Domain Grounding via Severity Bands 7

Decision Logic and Behavior **7**

Anomaly Detection Logic 7

Incident Severity Classification 8

Safety, Reliability, and Transparency **8**

Context Window Control and Latency Optimization 8

System Observability and Analytics 8

Observed Behavior and Limitations **10**

Experimental Evaluation and Sample Results 10

LTM-Driven Behavioral Adaptation and Inconsistencies 10

Ethical and Responsible Use Considerations **10**

Future Improvements **16**

References **16**

List of Figures

1 Workflow Graph. 5

2 Severity Bands. 7

3 System Analytics: tool, token, and function usage. 9

4 Sample test-case for anomaly `CLASS_0`. 11

5 Sample test-case for anomaly `CLASS_1`. 12

6 Sample test-case for anomaly `CLASS_2`. 13

7 Sample test-case (Asset 3, Test 3) for anomaly `CLASS_1` without LTM. . . . 14

8 Sample test-case (Asset 3, Test 3) for anomaly `CLASS_1` using LTM. 15

9 Repeat after several days: Sample test-case (Asset 3, Test 3) using LTM. . . 15

Overview

We developed an agentic workflow (Plaat et al., 2025; Hughes et al., 2025) to classify pre-detected anomalies in industrial telemetry into three distinct operational response classes. The core system goal is to automate incident triage by identifying failure modes, conducting root-cause analysis, recommending corrective actions, and routing incidents to the appropriate operational tier for execution.

Built using LangGraph (LangChain Inc., 2026), the orchestration framework utilizes a single central Large Language Model (LLM) (Wikipedia contributors, 2026b) acting as a ReAct (Reasoning and Acting) agent (Yao et al., 2023) within the classifier node to manage workflow decisions.

Monitored Telemetry Parameters

The system evaluates four core telemetry parameters representative of critical industrial assets, such as a Multistage Centrifugal Water Pump (Wikipedia contributors, 2026a):

1. **Temperature (`temperature_c`):** Monitored at bearings, lubrication loops, or process fluids to track thermal friction and cooling efficiency.

2. **Vibration (`vibration_hz`):** Tracks mechanical oscillations to detect shaft misalignment, rotor imbalance, or bearing degradation.

3. **Pressure (`pressure_psi`):** Measures internal suction, discharge, or hydraulic line pressures.

4. **Power (`power_kw`):** Tracks real-time electrical power consumption to evaluate motor load and overall efficiency.

System Architecture and Design

Workflow Graph

The agentic system's execution flow is modeled as a directed orchestration graph shown in Fig. 1.

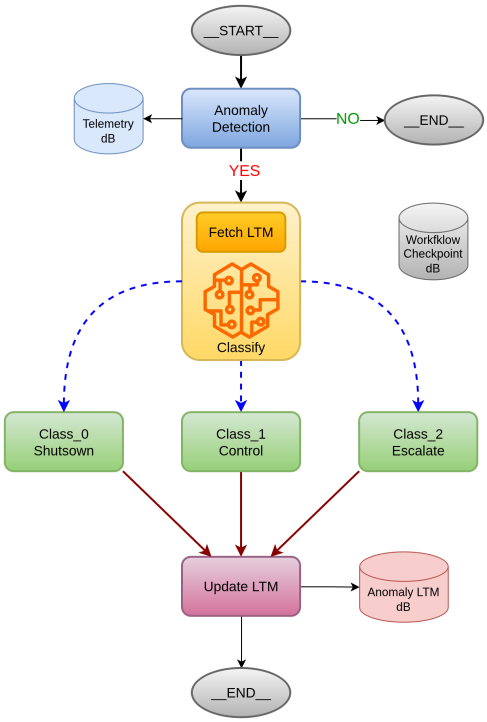


Figure 1

Workflow Graph.

The execution sequence follows a multi-stage routing pipeline:

1. **Anomaly Detector Node:** Telemetry data enters the anomaly detector node and is logged to the telemetry database. If no anomaly is detected, the workflow terminates. If an anomaly is present, data is forwarded to the Classifier Agent.

2. **Classifier Agent Node:** Operates as a ReAct agent using few-shot prompting (Gadesha, 2026) with clearly defined response classes. Context is enriched using the 3 most recent anomaly records retrieved from Long-Term Memory

(LTM) (Stryker, 2026; Chugani, 2025). The agent utilizes a single tool to fetch standardized severity bands for the current asset.

3. **`CLASS_0` Node:** Executes immediate asset shutdown protocols for critical safety hazards.

4. **`CLASS_1` Node:** Triggers automated control adjustments (e.g., via Variable Frequency Drive) for controllable process drifts.

5. **`CLASS_2` Node:** Escalate complex or unhandled anomalies by generating a comprehensive report for an on-site engineer.

6. **LTM Update Node:** Updates the Long-Term Memory database with event details across all anomaly paths.

Data Storage and Persistence Architecture

To maintain zero external deployment overhead while preserving operational isolation, the system employs three light-weight SQLite databases:

- **Telemetry Database:** Stores continuous time-series logging of all incoming asset telemetry.
- **Short-Term Memory (STM) Checkpoint Store:** Captures full state checkpoints for offline auditing, system analytics, and performance tracking.
- **Long-Term Memory (LTM) Store:** Maintains a historical record of past anomaly events and resolutions per asset to provide historical context to the agent.
- **System Logging:** The system logs (error, debug, warning, info) are saved in **JSONL** format for easy offline parsing.

Please see sample of log and databases in the notebook.

Simplified Domain Grounding via Severity Bands

While a production system could integrate Retrieval-Augmented Generation (RAG) (Wikipedia contributors, 2026c) against technical manuals, this implementation uses a deterministic threshold tool to convert raw numerical anomalies into standardized severity bands (*Critically Low, Low, Normal, High, Critically High*), as shown in Fig. 2.

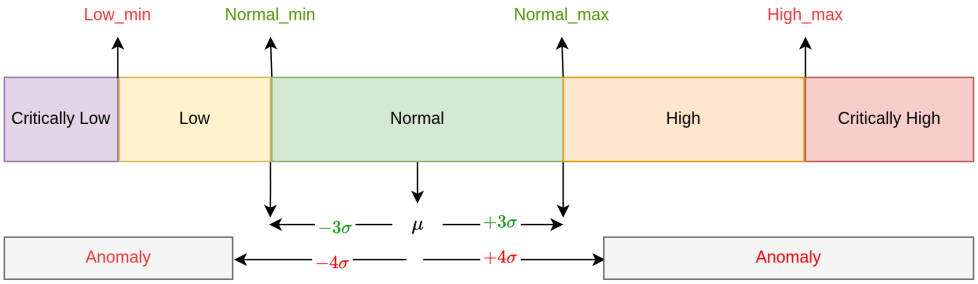


Figure 2

Severity Bands.

Bypassing raw statistical metrics (e.g., standard deviations and means) inside the LLM prompt prevents hallucination and reasoning errors. This abstraction aligns with practical engineering realities where a deviation of 10σ might represent a minor operational shift in one parameter but an immediate catastrophic failure in another.

Decision Logic and Behavior

Anomaly Detection Logic

Rather than relying on complex machine learning models, the anomaly detection node utilizes a transparent statistical rule set defined around baseline Gaussian distributions:

- The full normal operating range spans $\pm 3\sigma$ centered around the mean.
- An anomaly is flagged whenever any metric exceeds $\pm 4\sigma$.
- The ranges $[-4\sigma, -3\sigma]$ and $[+3\sigma, +4\sigma]$ serve as operational tolerance buffers prior to reporting an anomaly.

Incident Severity Classification

Once an anomaly is flagged, the central classifier agent categorizes the incident into one of three operational classes:

0. `CLASS_0` **(Immediate Shutdown):** Triggered by severe threshold breaches (Critically High) on any dangerous parameter (temperature, pressure, or vibration) requiring instant asset shutdown to prevent catastrophic damage.

1. `CLASS_1` **(Automated Control):** Triggered by minor single-parameter operational stress or controllable process drift solvable via automated actuators (e.g., VFD).

2. `CLASS_2` **(Escalate to Onsite Engineer):** Triggered by physical degradation or complex multi-telemetry anomalies requiring physical inspection, maintenance, or manual intervention.

Safety, Reliability, and Transparency

Context Window Control and Latency Optimization

Response latency is critical in industrial automation. To minimize inference delay, LTM access is kept outside the agent's tool execution loop; instead, relevant LTM context (the last 3 historical anomalies) is directly pre-loaded into the prompt before invocation.

Furthermore, because state checkpoints in STM are reserved strictly for offline auditing and analytics, input context lengths remain deterministic. Complex context management techniques—such as sliding-window truncation or message summarization—are unnecessary, preventing context drop and ensuring consistent reasoning across runs.

System Observability and Analytics

The graph checkpointing mechanism captures system state after every node transition, enabling thorough post-hoc auditing. Figures 3 illustrate the offline tracking metrics extracted from execution logs, covering LLM tool call frequencies, cumulative token consumption, and node execution profiles.

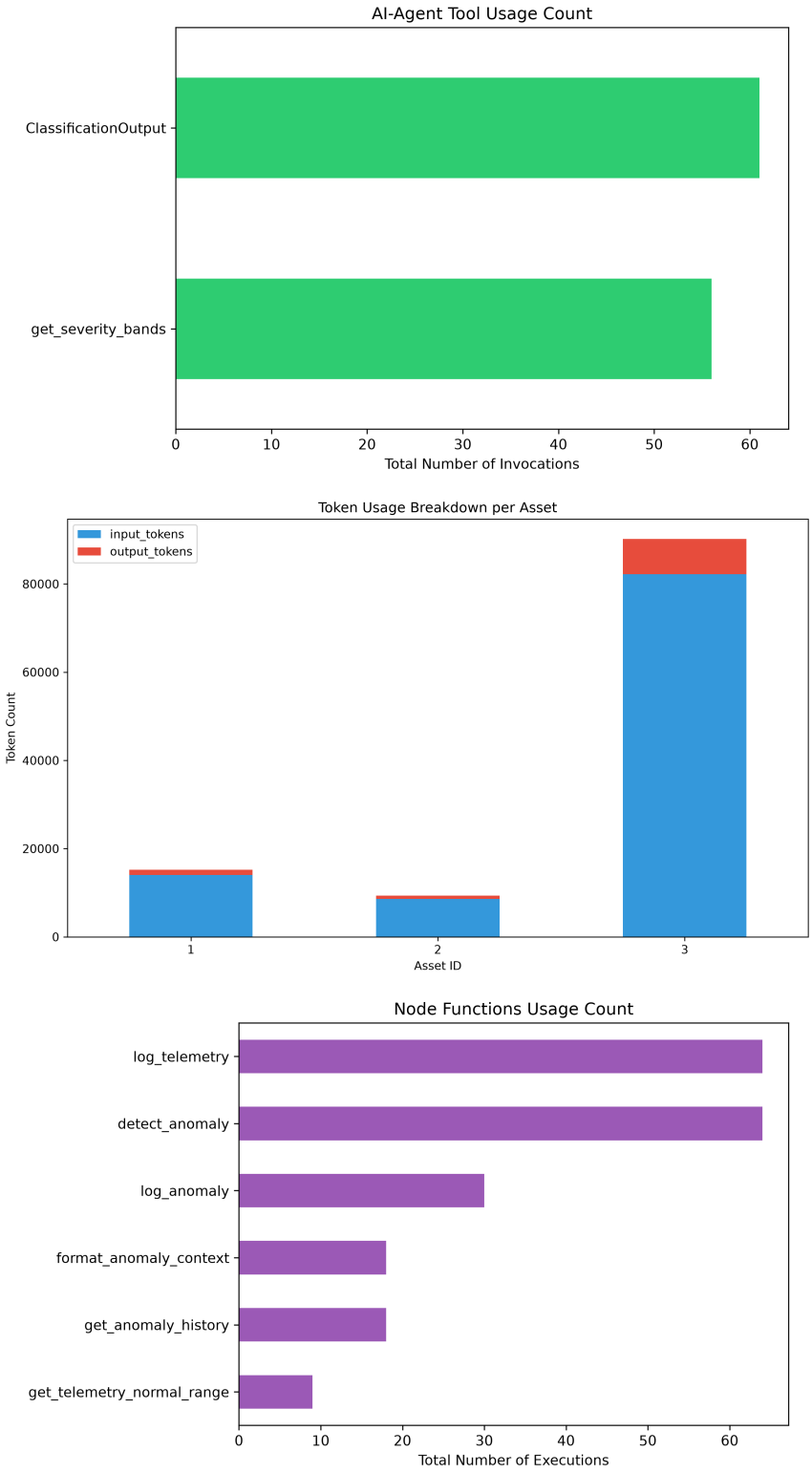


Figure 3

System Analytics: tool, token, and function usage.

Observed Behavior and Limitations

Experimental Evaluation and Sample Results

The framework was evaluated across 3 simulated centrifugal pump assets across 9 total test scenarios (3 anomaly cases per asset). Telemetry trends were simulated using sigmoid-like curve transitions for realistic parameter drift. Across baseline runs, the system executed expected actions across all classes, as illustrated in Fig. 4 (`CLASS_0`), Fig. 5 (`CLASS_1`), and Fig. 6 (`CLASS_2`).

LTM-Driven Behavioral Adaptation and Inconsistencies

To assess the impact of Long-Term Memory, a `CLASS_1` candidate scenario was evaluated under different LTM states:

- **Without LTM:** The agent consistently routed the event to `CLASS_1` for automated adjustment (Fig. 7).
- **With LTM (Identical Past Failures Loaded):** Upon observing in LTM that prior automated adjustments had failed to clear the anomaly, the agent adapted its behavior and escalated the event to `CLASS_2` (Fig. 8).
- **Repeated Run After Time Delay:** In subsequent runs, the agent attempted an automated control intervention (`CLASS_1`) once before re-escalating to `CLASS_2` on the next occurrence (Fig. 9).

While this shift demonstrates adaptive decision-making (giving automated control a secondary trial before escalating), it also highlights potential non-deterministic reasoning in LLM-based controllers.

Ethical and Responsible Use Considerations

Automating control decisions and emergency shutdowns in industrial physical infrastructure introduces significant operational risks. Granting autonomous agents direct execution authority over `CLASS_0` emergency shutdowns poses a trade-off:

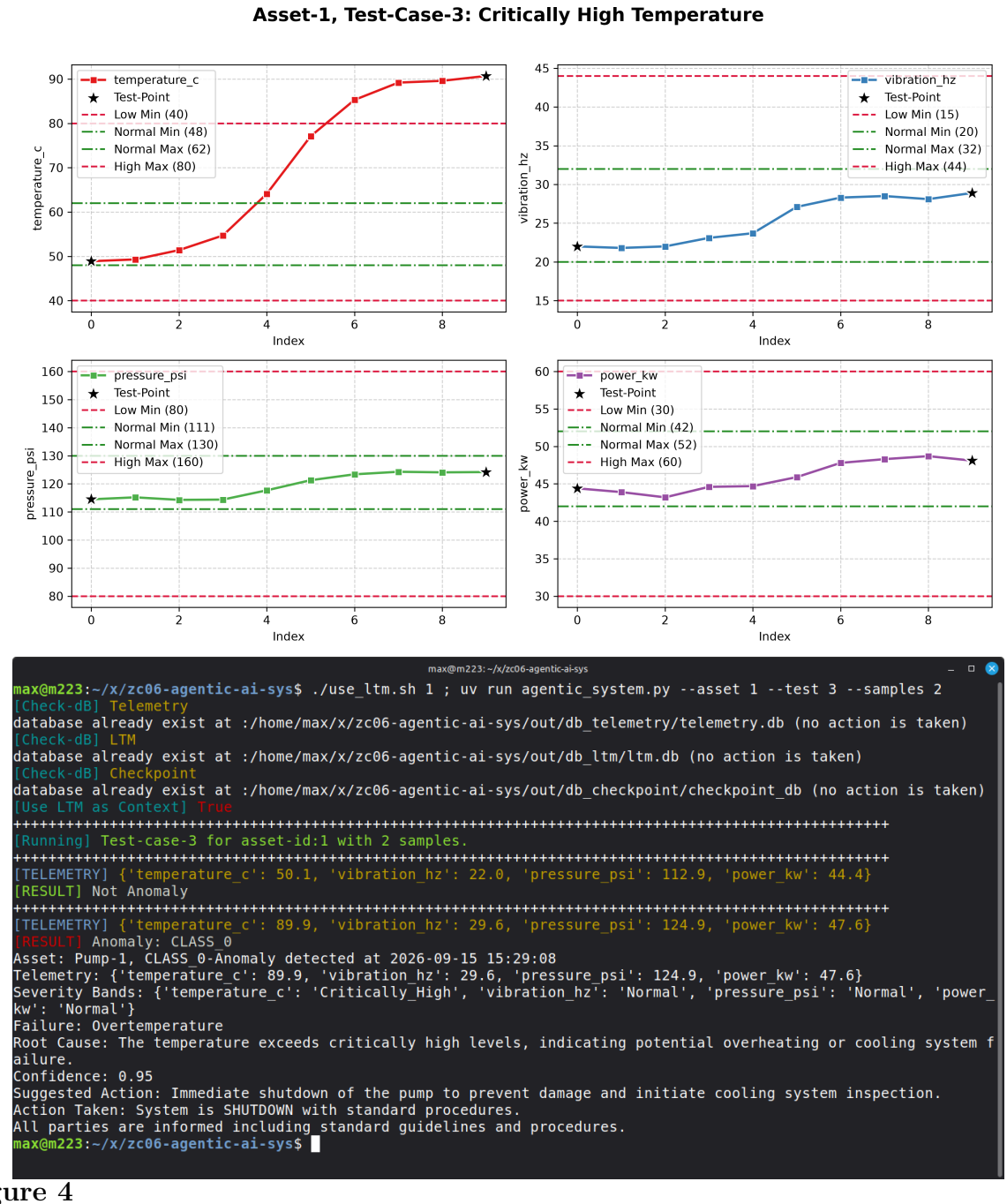
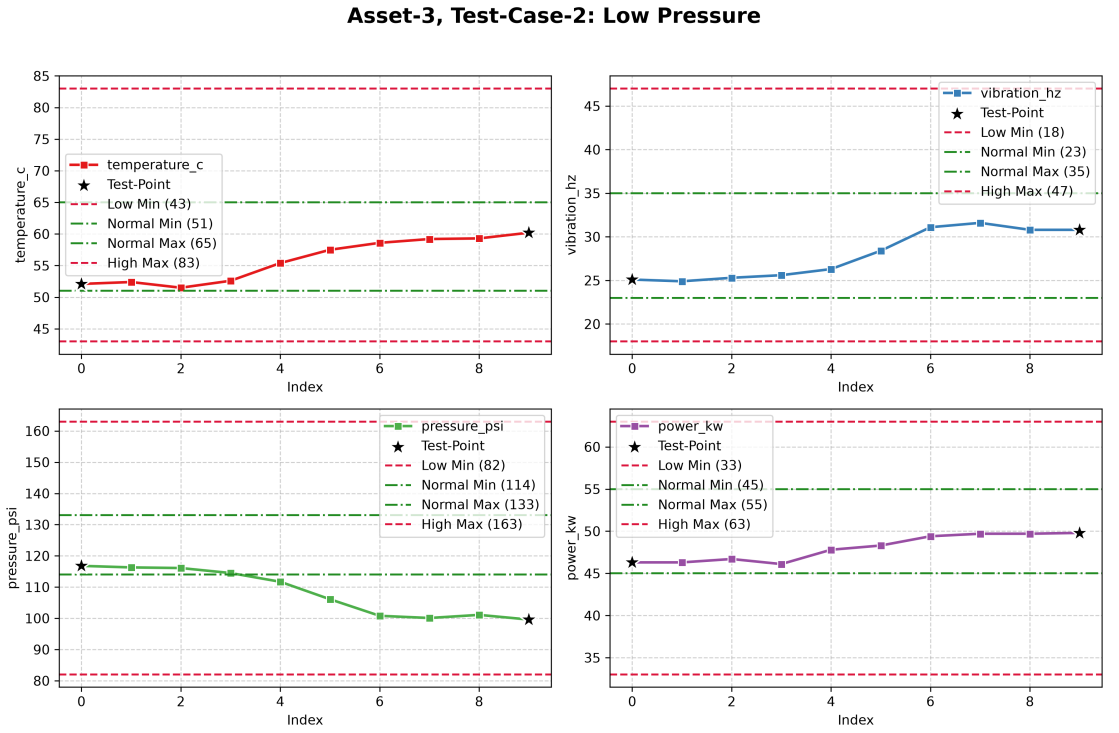


Figure 4

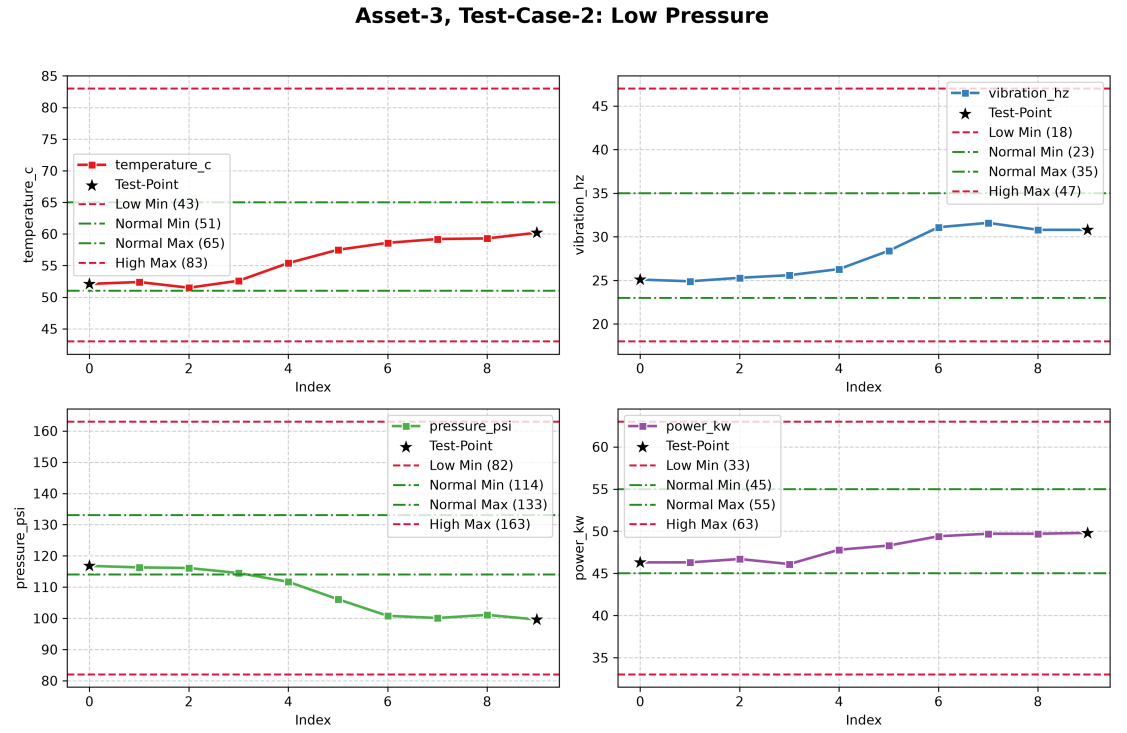
Sample test-case for anomaly `CLASS_0`.




```
max@m223:~/x/zc06-agentic-ai-sys
Updated 1 path from the index
max@m223:~/x/zc06-agentic-ai-sys$ ./use_ltm.sh 1 ; uv run agentic_system.py --asset 3 --test 2 --samples 2
[Check-dB] Telemetry
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_telemetry/telemetry.db (no action is taken)
[Check-dB] LTM
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_ltm/ltm.db (no action is taken)
[Check-dB] Checkpoint
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_checkpoint/checkpoint_db (no action is taken)
[Use LTM as Context] True
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[Running] Test-case-2 for asset-id:3 with 2 samples.
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[TELEMETRY] {'temperature_c': 52.1, 'vibration_hz': 25.0, 'pressure_psi': 115.9, 'power_kw': 46.4}
[RESULT] Not Anomaly
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[TELEMETRY] {'temperature_c': 59.9, 'vibration_hz': 31.6, 'pressure_psi': 99.9, 'power_kw': 49.6}
[RESULT] Anomaly: CLASS_1
Asset: Pump-3, CLASS_1-Anomaly detected at 2026-09-15 16:00:06
Telemetry: {'temperature_c': 59.9, 'vibration_hz': 31.6, 'pressure_psi': 99.9, 'power_kw': 49.6}
Severity Bands: {'temperature_c': 'Normal', 'vibration_hz': 'Normal', 'pressure_psi': 'Low', 'power_kw': 'Normal
'}
Failure: Normal Operation
Root Cause: All parameters are within normal operational ranges, indicating normal operation.
Confidence: 1.0
Suggested Action: No immediate action required. Continue monitoring.
Action Taken: set VFD control to increase pressure_psi
from 99.9-Low pressure to 123.5 (mean Normal value).
max@m223:~/x/zc06-agentic-ai-sys$
```

Figure 5

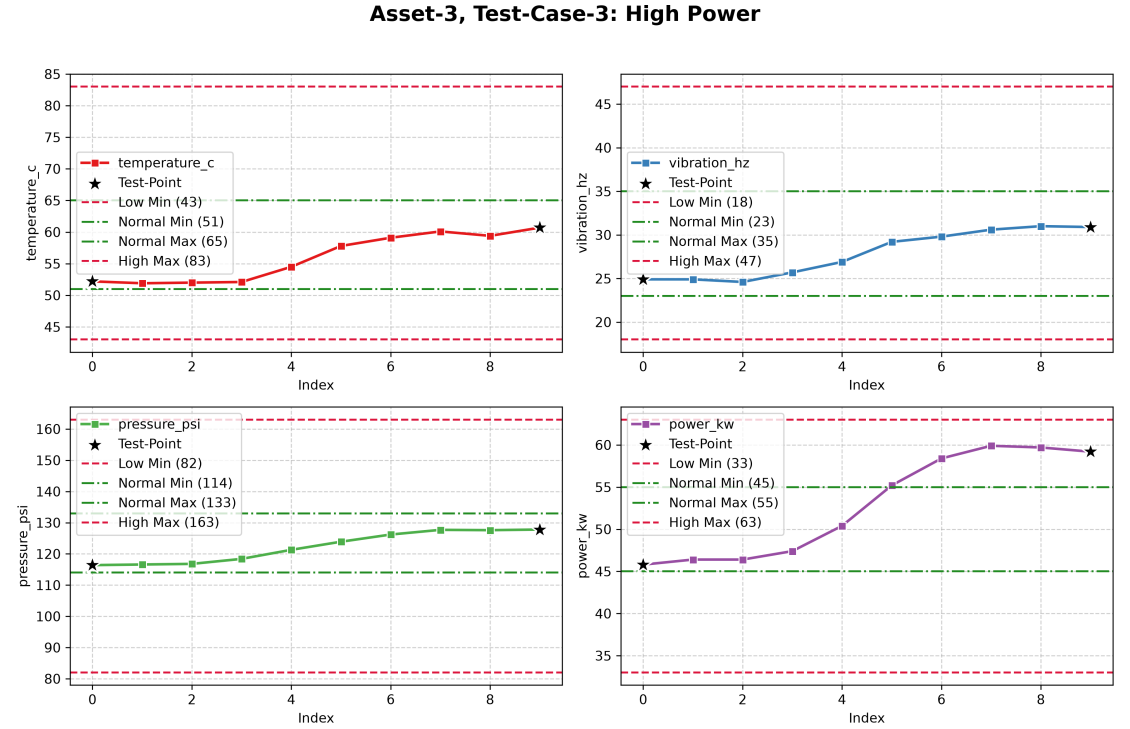
Sample test-case for anomaly `CLASS_1`.




```
max@m223:~/x/zc06-agentic-ai-sys
Updated 1 path from the index
max@m223:~/x/zc06-agentic-ai-sys$ ./use_ltm.sh 1 ; uv run agentic_system.py --asset 3 --test 2 --samples 2
[Check-dB] Telemetry
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_telemetry/telemetry.db (no action is taken)
[Check-dB] LTM
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_ltm/ltm.db (no action is taken)
[Check-dB] Checkpoint
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_checkpoint/checkpoint_db (no action is taken)
[Use LTM as Context] True
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[Running] Test-case-2 for asset-id:3 with 2 samples.
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[TELEMETRY] {'temperature_c': 52.1, 'vibration_hz': 25.0, 'pressure_psi': 115.9, 'power_kw': 46.4}
[RESULT] Not Anomaly
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[TELEMETRY] {'temperature_c': 59.9, 'vibration_hz': 31.6, 'pressure_psi': 99.9, 'power_kw': 49.6}
[RESULT] Anomaly: CLASS_1
Asset: Pump-3, CLASS_1-Anomaly detected at 2026-09-15 16:00:06
Telemetry: {'temperature_c': 59.9, 'vibration_hz': 31.6, 'pressure_psi': 99.9, 'power_kw': 49.6}
Severity Bands: {'temperature_c': 'Normal', 'vibration_hz': 'Normal', 'pressure_psi': 'Low', 'power_kw': 'Normal
'}
Failure: Normal Operation
Root Cause: All parameters are within normal operational ranges, indicating normal operation.
Confidence: 1.0
Suggested Action: No immediate action required. Continue monitoring.
Action Taken: set VFD control to increase pressure_psi
from 99.9-Low pressure to 123.5 (mean Normal value).
max@m223:~/x/zc06-agentic-ai-sys$
```

Figure 6

Sample test-case for anomaly `CLASS_2`.




```
max@m223:~/x/zc06-agentic-ai-sys
max@m223:~/x/zc06-agentic-ai-sys$ ./use_ltm.sh 0 ; uv run agentic_system.py --asset 3 --test 3 --samples 2
[Check-dB] Telemetry
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_telemetry/telemetry.db (no action is taken)
[Check-dB] LTM
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_ltm/ltm.db (no action is taken)
[Check-dB] Checkpoint
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_checkpoint/checkpoint_db (no action is taken)
[Use LTM as Context] False
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[Running] Test-case-3 for asset-id:3 with 2 samples.
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[TELEMETRY] {'temperature_c': 52.1, 'vibration_hz': 25.0, 'pressure_psi': 115.9, 'power_kw': 46.4}
[RESULT] Not Anomaly
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[TELEMETRY] {'temperature_c': 59.9, 'vibration_hz': 31.6, 'pressure_psi': 127.9, 'power_kw': 59.6}
[RESULT] Anomaly: CLASS_1
Asset: Pump-3, CLASS_1-Anomaly detected at 2026-09-15 16:14:40
Telemetry: {'temperature_c': 59.9, 'vibration_hz': 31.6, 'pressure_psi': 127.9, 'power_kw': 59.6}
Severity Bands: {'temperature_c': 'Normal', 'vibration_hz': 'Normal', 'pressure_psi': 'Normal', 'power_kw': 'Hig
h'}
Failure: High Power Consumption
Root Cause: The power consumption is higher than normal, indicating potential overload or inefficiency in the pu
mp system.
Confidence: 1.0
Suggested Action: Monitor the pump closely, check for any abnormal operational conditions, and consider reducing
 load if necessary.
Action Taken: set VFD control to decrease power_kw
from 59.6-High power_kw to 50.0 (mean Normal value).
max@m223:~/x/zc06-agentic-ai-sys$
```

Figure 7

Sample test-case (Asset 3, Test 3) for anomaly `CLASS_1` *without LTM.*

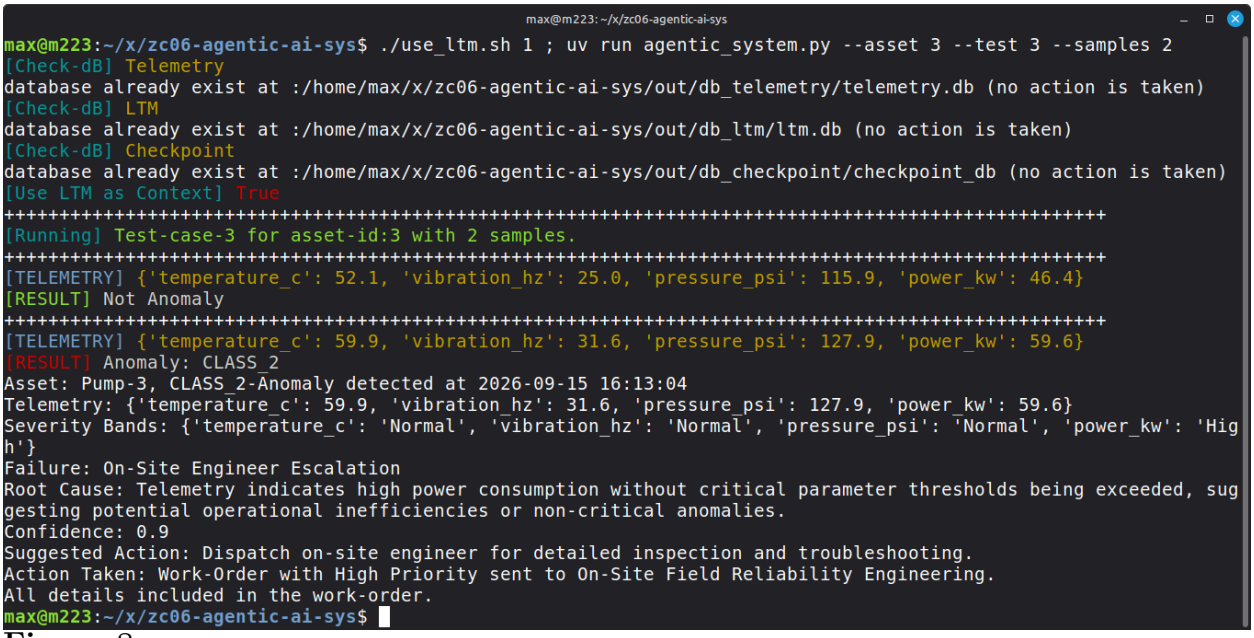


Figure 8

Sample test-case (Asset 3, Test 3) for anomaly ***CLASS_1*** *using LTM.*

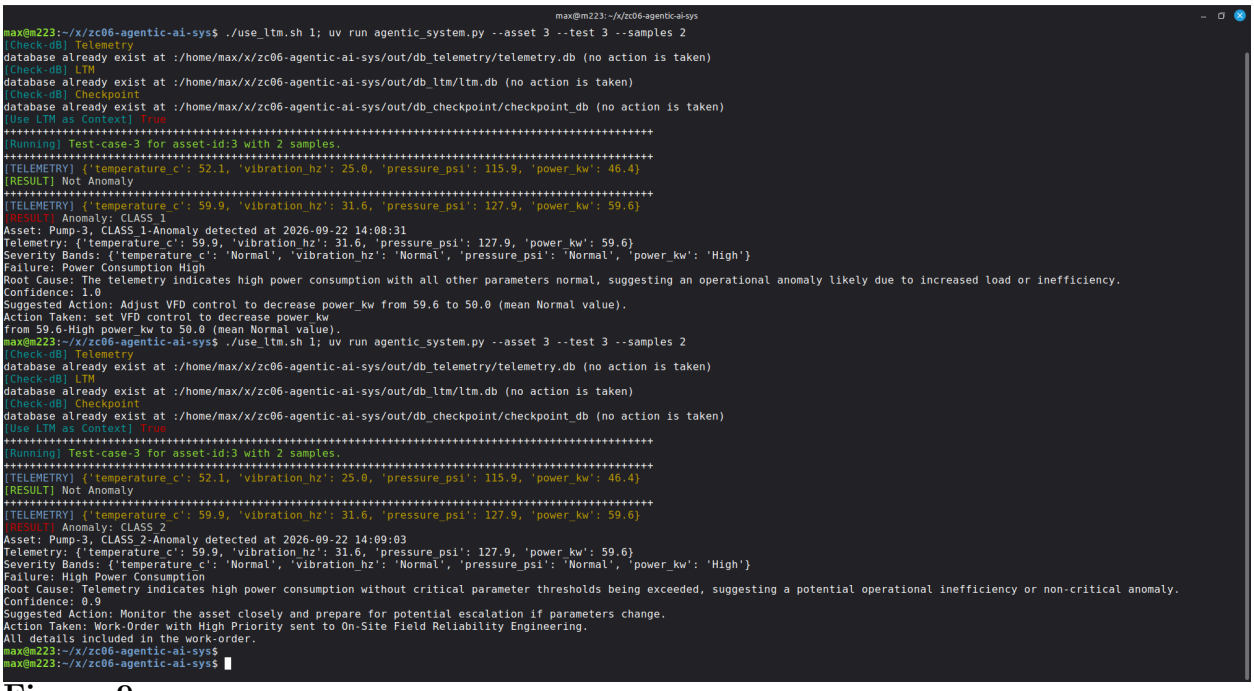


Figure 9

Repeat after several days: Sample test-case (Asset 3, Test 3) using LTM.

- **Safety vs. Financial Cost:** An automated emergency shutdown prevents catastrophic physical failure, protecting personnel and infrastructure. However, false positives cause expensive, unnecessary production halts.
- **Accountability and Delays:** Requiring a Human-in-the-Loop (HITL) for critical shutdowns mitigates false-positive costs but introduces latency that may lead to physical destruction during fast-moving failure modes.

Future Improvements

To address present system limitations, future development will target the following areas:

1. **Machine Learning Anomaly Detection:** Replace rule-based deviation bounds with trained machine learning models fitted on historical industrial telemetry.
2. **Generative Data Pipelines:** Integrate Generative AI models to synthesize diverse normal and rare failure-mode telemetry for robust benchmark testing.
3. **Expanded Actuator Integration:** Extend control capability beyond Variable Frequency Drives (VFDs) to include actuators like bypass and recirculation valves.
4. **Closed-Loop Simulation:** Implement continuous closed-loop feedback environments to dynamically evaluate asset recovery following agent-initiated control adjustments.
5. **Configurable Safety Guards:** Incorporate optional Human-in-the-Loop approval nodes for emergency shutdowns based on facility-specific risk profiles.

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